

CHE 425

1. Course number and name

CHE 425 Nanotechnology for Pharmaceutical Applications

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Ying Liu

4. Text book, title, author, publisher, year

Drug Delivery: Engineering Principles for Drug Therapy Topics in Chemical Engineering, W. Mark Saltzman, Oxford, 2001.

a. Other supplemental materials

Selected reading materials from high-impact journals and a few textbooks, including but not limited to, Science, Nature, JACS, PRL, Biomacromolecules, ACS Nano, Nano Letters, Journal of Colloids and Interfacial Science.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Emerging role of nanostructures in drug development and delivery. Principles of nanostructure formation, characterization, surface functionalization, in vitro and in vivo transport, and visualization.

b. Prerequisites or co-requisites

CHE 301 and CHE 311 and CHE 312; or consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will be able to:

- Understand the fundamental and essential concepts used in nanotechnology, synthesis, fabrication and characterization.
- Analyze thermodynamic and kinetic control of the nanoparticle formation and stabilization
- Analyze drug loading, encapsulation efficiency, and transport
- Discuss nanomaterial selection, design and fabrication.
- Discuss nanomaterial impact on the environment
- Apply written and oral communication skills to a final term paper and presentation

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3)	an ability to communicate effectively with a range of audiences.			X	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Factors that impact the development ability of drug candidates.
- Evolution of controlled drug delivery systems.
- Advanced Nanotechnologies and Micro technologies for Drug Delivery
- Self-assembled delivery vehicles: basic theoretical considerations and modeling concepts.
- Nanoparticle characterization
- Liposomes as intravenous solubilizers.
- Polymeric nanoparticle drug delivery systems.
- Nano suspensions with enhanced drug dissolution rates of poorly water-soluble drugs.
- Pharmaceutical Fundamentals
- Physiological, Biochemical, and Chemical barriers to Oral Drug Delivery.
- Cell culture models for drug transport studies.